Reg. No. | | | | | | | | | | [12EE8409]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

ENERGY STORAGE CONSERVATION AND AUDIT

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	Explain the need for energy storage. Mention the various methods of energy storage.	06
	b.	With the help of a neat sketch, explain the function of compressed air energy storage system.	08
2.	a.	Explain the different ways of storing energy using hydrogen.	07
	b.	Explain the importance of sensible heat and latent heat storage.	07
3.	a.	Mention the detail classification of burners.	04
	b.	List atleast six factors affecting the fuel economy and design of furnaces.	06
	c.	Explain the functions of the following boiler accessories. i. Economizer ii. Super heater.	04
4.	a.	What are the benefits of a cogeneration system?	06
	b.	With a neat sketch explain the back pressure system and extraction condensing system of steam turbine.	08
5.	a.	Explain the classification of thermal insulation.	06
	b.	List six properties of a refractory material.	06
	c.	Mention any four criteria used for the selection of refractories.	02
6.	a.	What is energy audit? How is energy audit classified?	04
	b.	Explain phase I, Phase II and Phase III energy audit in detail.	10
7.	a.	Write a note on material balance and energy balance.	08
	b.	List any six energy auditing instruments and mention its use.	06
8.	a.	Define the following terms: i. Load factor ii. Demand factor iii. Diversity factor	06

b. A domestic consumers monthly consumption of electricity is as follows: **08**

5 lamps of 100 watts each for 4 hours, a geyser of 2.5kW for one hour daily, room heater of 1kw for 2hours daily.

Calculate the monthly bill as per the following tariff:

Rs. 4/- per kwh for the first 25 units.

Rs. 5.50/- per kwh for the next 50 units.

Rs. 6.50/- per kwh for the remaining units.